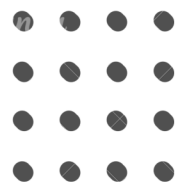


VMES Grinds

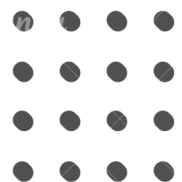




VMES Grinds

Pilot Program - Week 3

Sliding Windows and Two Pointers





Leetcode: 881. Boats to Save People





881. Boats to Save People

Solved

Medium

Topics

Companies

You are given an array `people` where `people[i]` is the weight of the i^{th} person, and an **infinite number of boats** where each boat can carry a maximum weight of `limit`. Each boat carries at most two people at the same time, provided the sum of the weight of those people is at most `limit`.

Return *the minimum number of boats to carry every given person*.



Example 1:

Input: people = [1,2], limit = 3

Output: 1

Explanation: 1 boat (1, 2)

Example 2:

Input: people = [3,2,2,1], limit = 3

Output: 3

Explanation: 3 boats (1, 2), (2) and (3)

Example 3:

Input: people = [3,5,3,4], limit = 5

Output: 4

Explanation: 4 boats (3), (3), (4), (5)



Constraints:

- $1 \leq \text{people.length} \leq 5 * 10^4$
- $1 \leq \text{people}[i] \leq \text{limit} \leq 3 * 10^4$



Explanation





Code





Code | Python3

```
class Solution:
    def numRescueBoats(self, people: List[int], limit: int) -> int:
        people.sort()
        l = 0
        r = len(people) - 1
        boat = 0
        while l <= r:
            remain = limit - people[r]
            r -= 1
            boat += 1
            if remain >= people[l]:
                l += 1
        return boat
```

^ View less



Time and Space Complexity Analysis





Code | Python3

Space - $O(n)$

```
class Solution:
    def numRescueBoats(self, people: List[int], limit: int) -> int:
        people.sort()
        l = 0
        r = len(people) - 1
        boat = 0
        while l <= r:
            remain = limit - people[r]
            r -= 1
            boat += 1
            if remain >= people[l]:
                l += 1
        return boat
```

Time - $O(n \log n)$

Time - $O(n)$

^ View less



Key Takeaway



Takeaway 1: If the problem asks us for
pairs, try considering sorting

Takeaway 2: Choose the local best →
leads to the optimal (Greedy Algorithm)

Takeaway 3: Greedy + Two Pointers
pattern